

Scenario Evaluation Worksheet

Instructions: Use this worksheet to document your analysis.

Part I: Analyze and document requirements. Map at least three requirements in each AI Product Development Stage.

Product Development Stages	Requirements
Ideation and Innovation Stage Requirements	Develop a clear problem statement and define the business case for creating the Smart Virtual Assistant.
	Identify and document core user needs (routine tasks to automate, pain points, expected assistant capabilities)..
	Conduct a detailed cost–benefit analysis showing expected productivity gains, resource savings, and measurable improvement metrics.
	Define key performance indicators (KPIs), including: • 100% adoption within 3 months • ≥4.5 satisfaction score • ≥98% response accuracy
	Identify required AI models (e.g., large language models, natural language understanding, classification systems). Assess integration needs with existing corporate systems (CRM, HRIS, internal tools, communication platforms). Establish cross-functional collaboration needs and identify domain experts. Document the Product Vision, Value Proposition, and success criteria.
Data Management Requirements	Identify and gather all data sources required to train and support the Virtual Assistant (FAQs, internal documents, knowledge bases, process workflows).
	Validate the quality, completeness, and credibility of all data to ensure reliable model outputs.
	Establish a robust data architecture with secure storage and retrieval systems aligned with company policies.
	Define data governance practices ensuring GDPR/CCPA compliance and internal privacy standards.
	Create a process for continuous data updates to maintain accuracy over time. Work with domain experts to ensure content is accurate, up to date, and relevant to user queries. Identify safeguards that ensure all data remains clean, bias-free, and securely accessible only to authorized personnel.

Research and Development Requirements	Build functional prototypes of the Smart Virtual Assistant to test core capabilities..
	Train, fine-tune, and validate AI models for accuracy, bias reduction, natural language understanding, and reliability.
	Develop test plans to validate user experience, system accuracy, and performance under varied conditions.
	Iterate based on user feedback—improve intent recognition, response quality, and conversational flow.
	Collaborate closely with design, engineering, and business teams to ensure alignment of technical and business goals. Identify blockers (technical limitations, integration gaps, performance issues) and create mitigation plans. Ensure the system can support at least 20,000 users initially and scale to 25,000 within one year.
Deployment Requirements	Conduct alpha and beta testing phases with real users across different company departments.
	Implement strong security controls to ensure safe authentication and authorized access only.
	Ensure reliability and minimize downtime by building high-availability infrastructure.
	Integrate the Virtual Assistant into existing corporate systems and communication platforms.
	Develop detailed onboarding resources and training materials to maximize user adoption. Monitor accuracy, satisfaction scores, usage trends, and system performance after launch. Regularly update the model and system with improvements, new features, and bug fixes. Create a long-term continuous improvement strategy with regular audits, data refresh cycles, and compliance checks.

Part II: Identify at least four roles required on the AI product management team. AI Product Manager is an essential role.

AI Product Team Role	Team Member Responsibilities
AI Product Manager	- Lead end-to-end product lifecycle, from concept to deployment and continuous improvement. - Define vision, KPIs, roadmap, and user requirements.

	• Coordinate cross-functional collaboration and align stakeholders. • Prioritize features and ensure the product meets business goals and user needs.
Data Scientist / ML Engineer	• Collect, clean, and process training data; evaluate model performance. • Build and fine-tune machine learning and NLU models. • Monitor accuracy, address drift, and optimize responses. • Ensure the model meets the 98% accuracy and bias-free requirements.
Software Engineer / Backend Developer	• Build the system infrastructure and integrate models with corporate systems. • Develop secure authentication, user management, and data flow pipelines. • Ensure system scalability to 20,000+ users. • Maintain backend reliability and performance.
UX/UI Designer	• Design user-friendly interfaces and conversational flows. • Improve usability, accessibility, and clarity of interactions. • Conduct user testing and gather feedback to refine user experience.
Content Specialist / Knowledge Manager	• Organize and maintain all knowledge sources used to feed the AI. • Ensure answers are accurate, updated, and aligned with company guidelines. • Write and maintain content templates and structured responses.
DevOps / IT Infrastructure Engineer	• Manage deployment environment, cloud resources, system monitoring, and uptime. • Ensure performance, security, and scalability requirements are met.

	Automate update processes and system maintenance tasks.

Part III: Identify at least four challenges or risks. Document your response to mitigate the challenges or risks.

Challenge or Risk	Mitigation Response
Data Quality and Quantity	Clean and structure all data before training. - Build processes to ensure data is updated regularly. - Validate all sources to avoid bias or inaccurate content.
Natural Language Understanding (NLU) Limitations	Fine-tune models using company-specific data. - Conduct extensive testing with users from different regions. - Add human-review workflows for ambiguous responses.
Scalability and Integration Challenges	- Ensure architecture supports 25,000 users. - Test system under heavy load and create fallback mechanisms. - Use APIs and modular design to integrate with existing corporate systems.
Privacy, Security, and Regulation Compliance	- Enforce strict authentication and role-based access. - Encrypt data in transit and at rest. - Conduct regular security audits; ensure GDPR/CCPA compliance.
User Adoption and Satisfaction Risks	- Provide user training, onboarding, and support. - Tailor responses to maximize usefulness and accuracy. - Use surveys to gather feedback and quickly resolve user concerns.
Maintenance and Model Drift	- Establish continuous monitoring of accuracy and user satisfaction. - Update content and retrain models regularly. - Set up alert systems for accuracy drop or drift.

Technical Downtime or Failure	• Build redundancy and backup systems. • Maintain 24/7 monitoring with rapid-response support. • Test failover scenarios prior to launch.
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